

MODULE 3: Data Link Layer, Ethernet & Wireless LANs

PCCST501 Computer Networks | Quick Exam Revision Guide | KTU S5 CSE | Prof. Anju Markose, VJCET

1. Data Link Control (DLC): Framing, Flow & Error Control

Framing Methods:

- **Character Count:** Specifies number of characters in header (single bit error desynchronizes all subsequent frames).
- **Byte Stuffing:** Uses FLAG byte (e.g. 0x7E) with ESC escape bytes when FLAG appears in data payload.
- **Bit Stuffing (HDLC):** Flag = 01111110. Sender inserts a '0' after every five consecutive '1's. Receiver strips the '0' following five '1's.

Key Concept / Exam Formula:

Cyclic Redundancy Check (CRC):

Given data bitstream M of length k and divisor polynomial G(x) of degree r:

1. Append r zeros to M ($M * 2^r$).
2. Perform Modulo-2 binary division (using XOR operations).
3. The r-bit remainder R is the Frame Check Sequence (FCS).
4. Transmit M followed by R. Receiver divides by G(x); if remainder is 0, frame is error-free.

2. Multiple Access Protocols (MAC)

- **Pure ALOHA:** Transmit immediately. Vulnerable period = $2 * T_{fr}$. Max throughput $S_{max} = 1/(2e) \approx 18.4\%$.
- **Slotted ALOHA:** Transmit only at slot boundaries. Vulnerable period = T_{fr} . Max throughput $S_{max} = 1/e \approx 36.8\%$.
- **CSMA/CD (Carrier Sense Multiple Access with Collision Detection):** Listen before and while transmitting. If collision occurs, transmit 32-bit jam signal and use **Binary Exponential Backoff** (wait $k * 512$ bit times, k in $[0, 2^n - 1]$).
- **Minimum Frame Size in CSMA/CD:** Frame transmission time must be at least $2 * \text{Propagation delay}$: $L_{min} = 2 * d_{prop} * \text{Bandwidth}$ (Standard Ethernet = 64 Bytes).

3. Link-Layer Addressing, Ethernet & Connecting Devices

- **MAC Address:** 48-bit (6 octets) globally unique physical address. First 24 bits = OUI (vendor), last 24 bits = NIC identifier.
- **ARP Protocol:** Broadcasts ARP Request (FF:FF:FF:FF:FF:FF) to find MAC of target IP. Destination replies with unicast ARP Reply.
- **Standard Ethernet Frame (IEEE 802.3):** Preamble (7B) + SFD (1B) + Dst MAC (6B) + Src MAC (6B) + Type (2B) + Payload (46-1500B) + CRC FCS (4B).
- **Learning Switches & VLANs:** Switches automatically build MAC tables by recording source MAC of incoming frames. VLANs (IEEE 802.1Q) partition a physical switch into multiple isolated broadcast domains.

4. Wireless LANs (IEEE 802.11) & Mobile IP

- **IEEE 802.11 Architecture:** Basic Service Set (BSS) → Extended Service Set (ESS) connected via Distribution System (DS).

- **CSMA/CA Protocol:** Uses DCF (Distributed Coordination Function) with SIFS < PIFS < DIFS inter-frame spaces and Network Allocation Vector (NAV).
- **Hidden & Exposed Node Problems:** Solved using optional RTS (Request to Send) / CTS (Clear to Send) 4-way handshake.
- **Mobile IP:** Mobile Node (MN) retains Home Address while assigning temporary Care-of Address (COA) via Foreign Agent (FA). Home Agent (HA) intercepts packets and tunnels them to FA (Triangle Routing).
- **Hands-on Raw Sockets:** `socket(AF_PACKET, SOCK_RAW, htons(ETH_P_ALL))` enables packet sniffing and frame injection at the link layer.